

**PROGRAM STUDI INFORMATIKA
FAKULTAS TEKNIK DAN INFORMATIKA
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IF350 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

Minggu/Pertemuan ke 1 – Software Engineering&Project Methodologies

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Weekly Learning Outcomes of the Course (Sub-CLO)



Sub-CPMK 0311

Explain the definition and scope of software engineering and the role of project management (C2)

Learning Objectives



1. Describe the software crisis and how the often dismal track record for information technology (IT) projects provides a motivation for changing how we view and manage IT projects.
2. Explain the socio-technical, project management and knowledge management approaches that support ITPM.
3. Define what an IT project is and describe its attributes.
4. Define the discipline called project management.
5. Describe the role and impact IT projects have on an organization.
6. Identify the different roles and interests of project stakeholders.
7. Describe the Software Engineering and project life cycle (PLC).
8. Describe the Systems Development Life Cycle (SDLC).
9. Describe the Waterfall method for developing the project's product or system.
10. Describe the Agile approach for developing the project's product or system as well as two commonly used approaches called eXtreme Programming (XP) and Scrum.
11. Describe and apply the concept of Learning Cycles and lessons learned.

Introduction

- Information Technology (IT) projects are organizational investments that require:
 - Time
 - Money
 - Other resources such as people, technology, facilities, etc.
- Organizations expect some type of value in return of this investment
- IT Project Management is a relatively new discipline that combines traditional Project Management with Software Engineering/Management Information Systems to make IT projects more successful.

An ITPM Approach

- Organizational resources are limited, so organizations must choose among competing interests to fund specific projects.
- This decision should be based on the value a competing project will provide to an organization.

Which Situation is Worse?

- Successfully building and implementing a system that provides little or no value to the organization.

OR

- Failing to implement an information system that could have provided value to the organization, but was poorly developed or poorly managed.

The Software Crisis

The CHAOS study published in 1995 by The Standish Group found that although the U.S spent over \$250 billion on IT projects, approximately

- 31% were cancelled before completion
- 53% were completed but over budget, over schedule, and did not meet original specifications.
- For mid-size companies, average cost overruns were 182%, while average schedule overruns were 202%!

The Newest Software Crisis

According to a recent article titled *“The AI Speed Trap: Why Software Quality Is Falling Behind in the Race to Release,”* the following alarming trends characterize the current software delivery environment:

- Two-thirds of businesses face a high risk of software outages in the coming year.
- Nearly 50% of organizations report losses of over \$1 million annually due to poor software quality.

This highlights a troubling trade-off:

- the push for speed
- especially through generative AI
- has outpaced safeguards like proper testing, resulting in outages, security risks, and mounting technical debt

Software Failures & Inefficiencies



The 2025 State of Software Engineering Excellence report, based on more than 650 engineering leaders, reveals critical bottlenecks in DevOps and developer workflows:

- 67% of teams cannot build and test their environments within 15 minutes.
- 64% still depend on manual steps for infrastructure code deployments.
- 61% report code reviews taking over a day.
- 55% lack quality gates in their build pipelines.
- 52% do not have essential incident management tools.
- 50% of application deployments are still manual.

These inefficiencies collectively result in millions lost annually due to wasted productivity, operational friction, and delayed innovation.

Summary of The CHAOS Study Results

<i>Company Size</i>	<i>Average Cost of Develop- ment</i>	<i>Average Cost Overruns</i>	<i>Average Schedule Overrun</i>	<i>Original Features and Functions Included</i>	<i>Successful Projects^a</i>	<i>Challenged Projects^b</i>	<i>Impaired Projects^c</i>
Large	\$2,322,000	178%	230%	42%	9%	61.5%	29.5%
Medium	\$1,331,000	182%	202%	65%	16.2%	46.7%	37.1%
Small	\$ 434,000	214%	239%	74%	28%	50.4%	21.6%

^a Completed on-time and on-budget

^b Completed, but over-budget, over schedule, and includes fewer features and functions than originally envisioned

^c Cancelled before completion

Why Do Projects Fail?

- Larger projects have the lowest success rate and appear to be more risky than medium and smaller projects:
Technology, business models and markets change so rapidly that a project that takes more than a year can be obsolete before they are completed.
- The Chaos study also provides some insight as to the factors that influence project success.

Summary of Factor Ranking for Successful, Challenged and Impaired Projects

<i>Rank</i>	<i>Factors for Successful Projects</i>	<i>Factors for Challenged Projects</i>	<i>Factors for Impaired Projects</i>
1	User involvement	Lack of user input	Incomplete requirements
2	Executive management support	Incomplete requirements	Lack of user involvement
3	Clear statement of requirements	Changing requirements & specifications	Lack of resources
4	Proper planning	Lack of executive support	Unrealistic expectations
5	Realistic expectations	Technology incompetence	Lack of executive support
6	Smaller project milestones	Lack of resources	Changing requirements specifications
7	Competent staff	Unrealistic expectations	Lack of planning
8	Ownership	Unclear objectives	Didn't need it any longer
9	Clear vision & objectives	Unrealistic time frames	Lack of IT management
10	Hard-working, focused team	New technology	Technology illiteracy

Has the Current State of IT Projects Changed Since 1995?

- The Standish Group has continued to study IT projects over the years.
- In general, IT Projects are showing higher success rates due to
 - Better project management tools & processes
 - Smaller projects
 - Improved communication among stakeholders
 - More skillful IT project managers

But there is still ample opportunity for improvement!

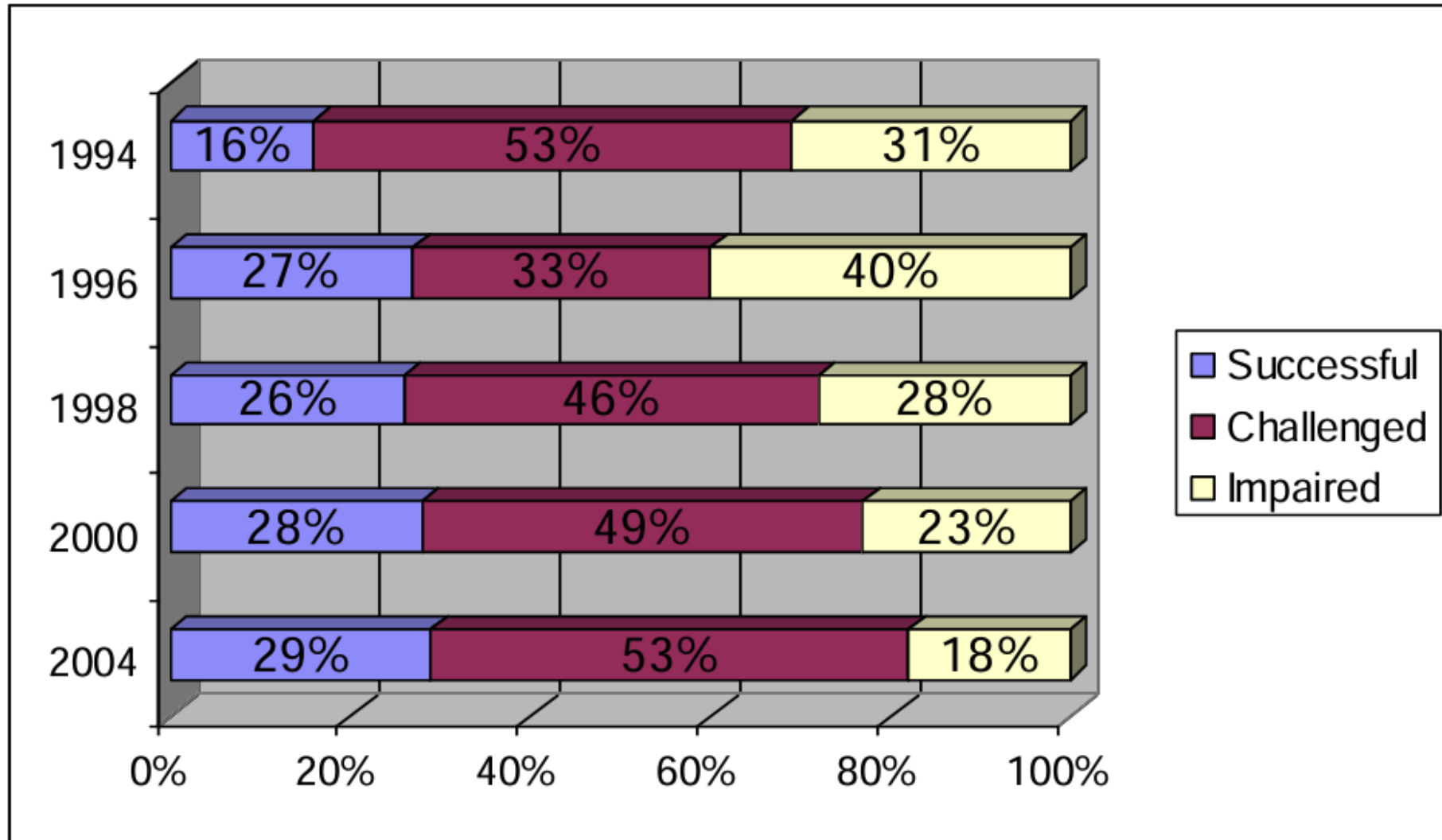
New Top Ten Factors for IT Project Success

Source: Extreme Chaos. The Standish Group International, Inc. 2001.
http://www.standishgroup.com/sample_research/index.php Vs 2020

Rank	Success Factor
1	Executive Support
2	User Involvement
3	Experienced Project Manager
4	Clear Business Objectives
5	Minimized Scope
6	Standard Software Infrastructure
7	Firm Basic Requirements
8	Formal Methodology
9	Reliable Estimates
10	Other

Rank	Success Factor	Description	
1	Executive Support	Strong backing from leadership.	
2	User Involvement	Collaborative input from users/stakeholders.	
3	Competent Staff	Skilled and experienced personnel.	
4	Smaller Milestones (Iterative)	Breaking projects into manageable increments.	
5	Clear Vision & Objectives	Well-defined goals aligned with business.	
6	Emotional Maturity / Soft Skills	Team and sponsorship emotional and behavioral competencies.	
7	Optimizing Scope	Controlled and focused project scope.	
8	Agile Process / PM Expertise	Agile methodology or high-level PM skills.	
9	Tools & Infrastructure	Reliable systems and supportive development environments.	
10	Good Team, Sponsor, Place	Holistic focus on people and environment (introduced in 2020).	

Summary of Chaos Studies



Improving the likelihood of Success

- Socio-technical Approach
- Project Management Approach
 - processes and infrastructure (Methodology)
 - Resources
 - Expectations
 - Competition
 - efficiency and effectiveness
- Project Management Approach
 - Knowledge Management Approach
 - lessons learned
 - best practices
 - shared knowledge

Project Management

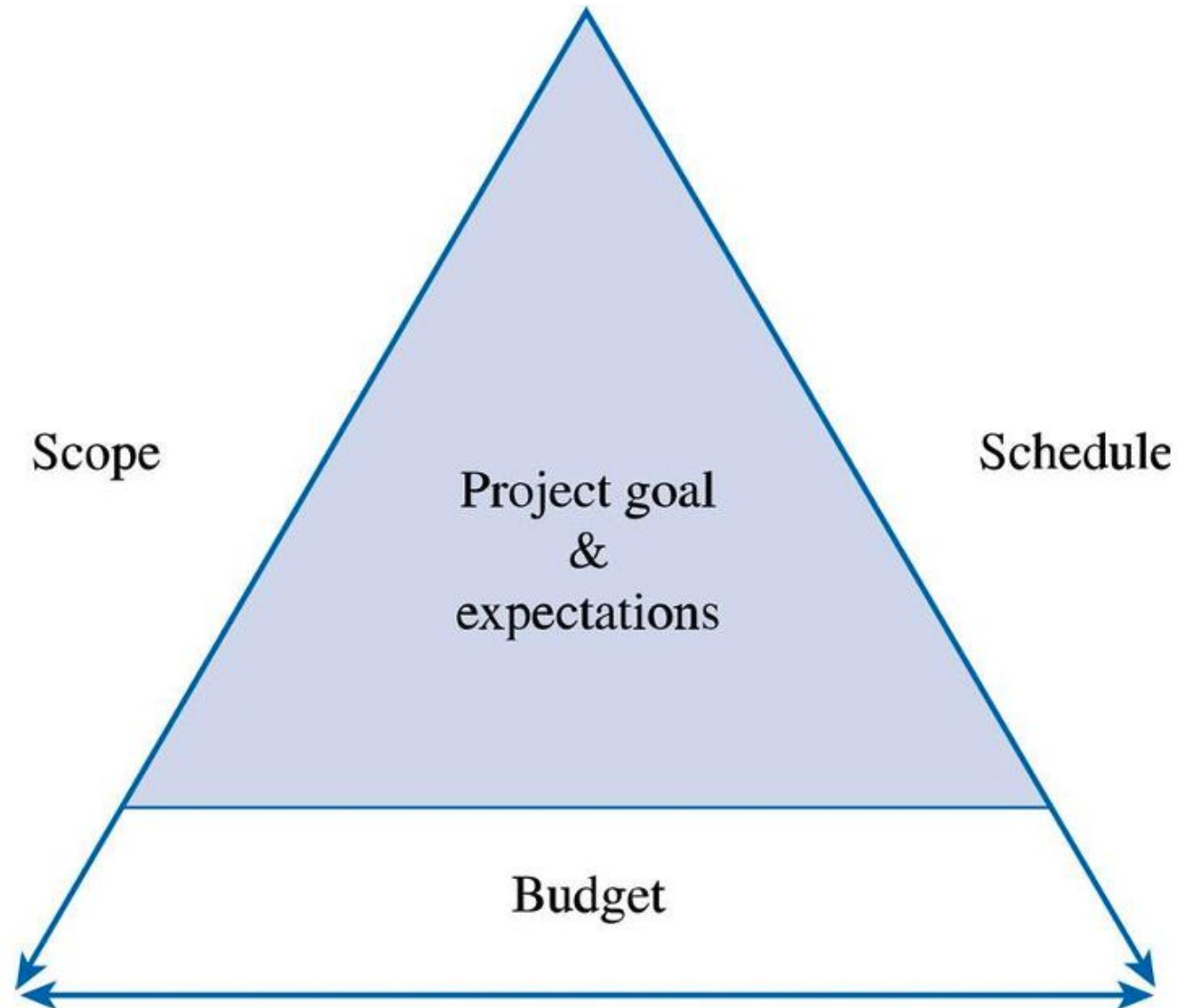
Definitions:

- A project is a temporary endeavor undertaken to accomplish a unique purpose.
- Project management is the application of knowledge, skills, tools, and techniques to project activities in order to meet or exceed project requirements

Project Attributes

- Time Frame
- Purpose (to provide value!)
- Ownership
- Resources (the triple constraint)
- Roles
 - Project Manager
 - Project Sponsor
 - SME (domain & technical)
- Risk & Assumptions
- Purpose (to provide value!)
- Ownership
- Interdependent Tasks
- Planned Organizational Change
- Operate in Environments Larger than the Project Itself

The Triple Constraint

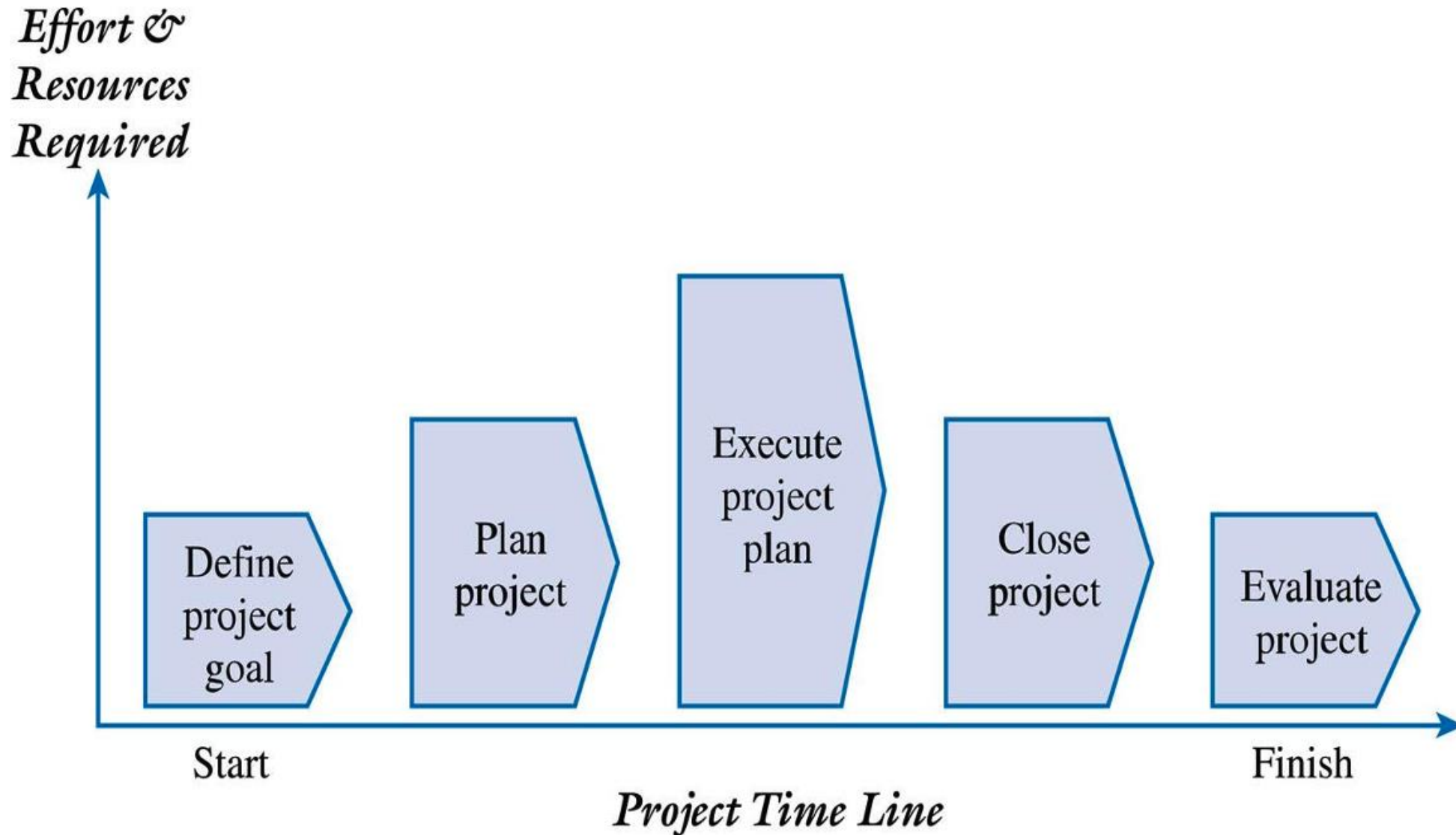


The Project Life Cycle and IT Development

Project Life Cycle (PLC)

- A collection of logical stages or phases that maps the life of a project from its beginning to its end in order to define, build and deliver the product of the project i.e., the information system
- Projects are divided into phases to increase manageability and reduce risk
 - Phase exits, stage gates, or kill points are decision points at the end of each phase to evaluate performance, correct problems or cancel the project
 - Fast tracking is the overlapping of phases to reduce the project's schedule, can be risky!

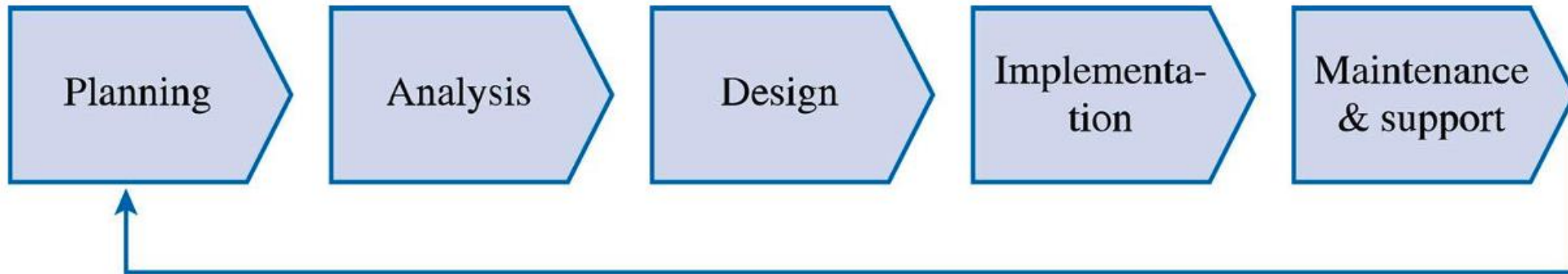
Generic Project Life Cycle



Systems Development Life Cycle (SDLC)

- Represents the sequential phases or stages an information system follows throughout its useful life
- Useful for understanding the development of the project's largest work product – the application system
- Phases/Stages
 - Planning
 - Analysis
 - Design
 - Implementation
 - Maintenance and Support

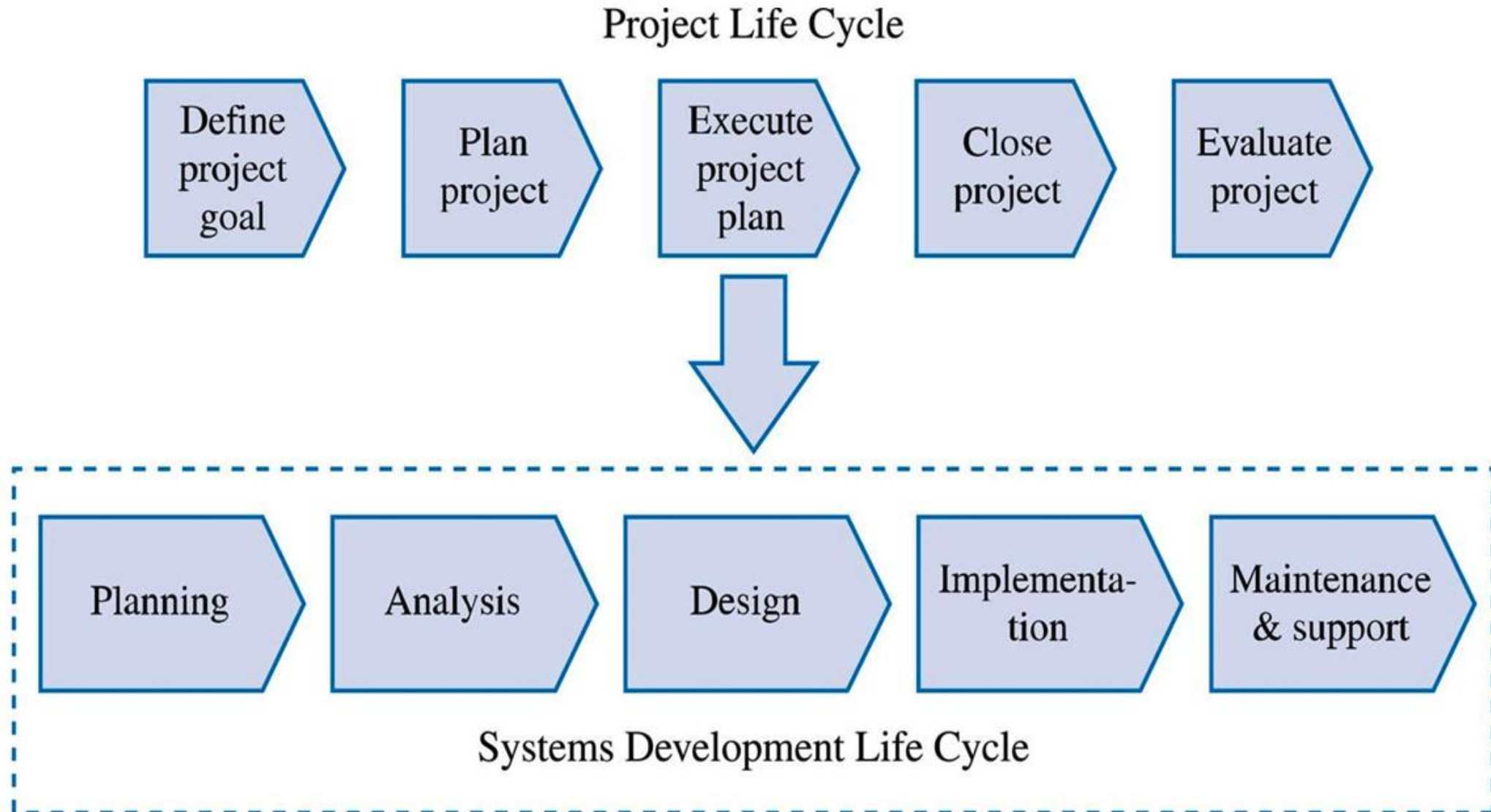
Systems Development Life Cycle (SDLC)



The Relationship Between the PLC and the SDLC

- The systems development life cycle (SDLC) becomes part of the project life cycle (PLC).
 - The PLC focuses on the project management phases, processes, tools and techniques for effectively managing the project.
 - The SDLC focuses on the software engineering phases, processes, tools and techniques for building and/or implementing the IT solution.

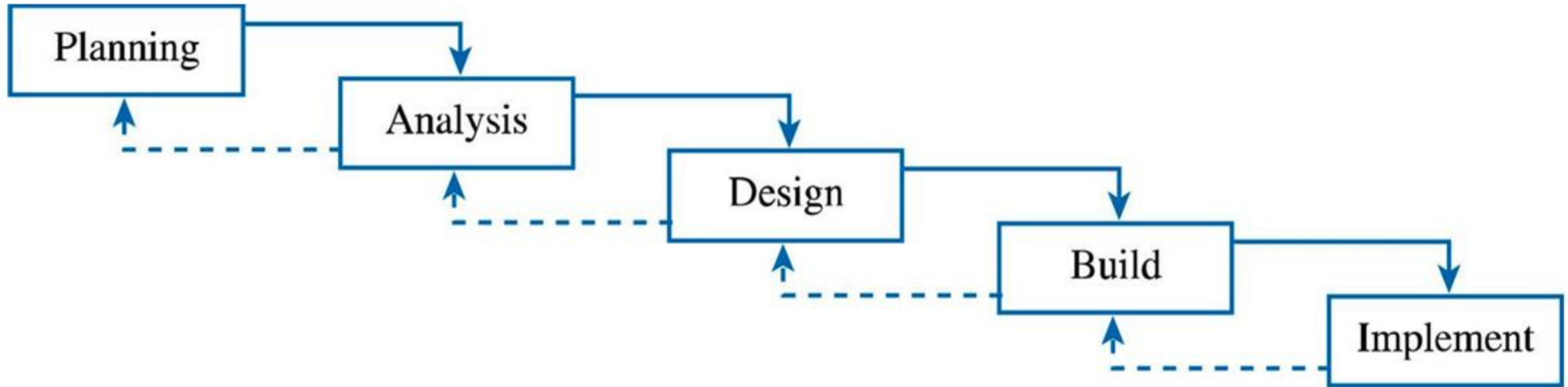
The Relationship Between the PLC and the SDLC



Putting the SDLC into Practice

- Structured Approach to Systems Development
 - Waterfall Method
- Rapid Applications Development (RAD)
 - Prototyping
 - Spiral Development
 - Extreme Programming

Structured Approaches Waterfall Method



Extreme Project Management (XPM)

- A new approach and philosophy to project management that is becoming increasingly popular.
- Characterizes many of today's projects that exemplify speed, uncertainty, changing requirements and high risks.
- Traditional project management often takes an orderly approach while XPM embraces the fact that projects are often chaotic and unpredictable.
- XPM focuses on flexibility, adaptability and innovation
- Traditional and new approaches together can provide us with a better understanding of how to improve the likelihood of project success.

Reference



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Visi

Menjadi Program Studi Strata Satu Informatika **unggulan** yang menghasilkan lulusan **berwawasan internasional** yang **kompeten** di bidang Ilmu Komputer (*Computer Science*), **berjiwa wirausaha** dan **berbudi pekerti luhur**.



Misi

1. Menyelenggarakan pembelajaran dengan teknologi dan kurikulum terbaik serta didukung tenaga pengajar profesional.
2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.